

Acid/Base Equilibria Tutorial

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Abstract

The concept of equilibrium as it applies to acids and bases in water is reviewed.

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Part I. Acid-Base Equilibria

1 Introduction

Acid and base equilibria are among the simpler concepts in freshman chemistry and yet many chemistry students often tell me that they only “got” the idea in their last year of university. I myself only figured out the practical use for acid equilibria calculations in graduate school. I hope to help you get a grasp on it in this course if you haven’t done so already.

I find it best to not dwell on the abstract idea of molecules dissociating and focus on the math. Please review your freshman chemistry notes and read this tutorial carefully if you are having any difficulty with acid equilibria. Practice the math described in this tutorial. Please ask questions if you still have difficulty with this concept.

Acid equilibrium is a very important concept in biochemistry. All biochemistry systems operate in an aqueous environment and involve interactions between molecules that may have acidic or basic groups. Are these groups in their acidic or their basic forms and are they positively charged, negatively charged or neutral under the reaction conditions.

We will need to understand acid equilibrium to know if these groups are in their acidic or basic forms.

Assumed Knowledge

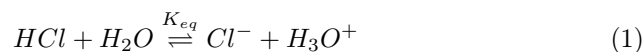
This tutorial assumes the following...

1. That you understand high school math and chemistry. You will especially want to review the mathematics of logarithms.

2 Strong Acids

The Brönsted concept of acids is that an acid is a group that can donate a proton to another group. A base is a group that can accept a proton.

The most common acid/base equilibrium is between an acid or a base and water. For example, let us consider the equilibrium between HCl, a strong acid, and H₂O. In this reaction HCl is donating a proton and H₂O is accepting it. HCl is the acid and H₂O is the base in this case.



The equation above includes H_2O . The concentration of the solvent does not change as it is in enormously greater concentration than any solute in any reasonable situation. So let us disregard the water and write the equilibrium as...



Writing the acid equilibrium this way is called acid dissociation. The equation is written as if the HCl was dissociating into chloride ion and proton. In fact the proton is transferred to water, which we are ignoring. The equilibrium constant for acid dissociation reactions is termed K_a .

$$K_a = \frac{[H^+][Cl^-]}{[HCl]} \quad (3)$$

The value of K_a for HCl has been measured to be $10^6 M$. The equilibrium results in virtually 100% of the HCl being dissociated. HCl is a strong acid because it dissociates completely in water. The concentration of proton (acid) in water is equal to the concentration of HCl that was added.

NaOH is a strong base. In water, it dissociates completely and the concentration of hydroxide ion (base) in water is equal to the concentration of NaOH added.

3 Dissociation of Water

H_2O is an acid as well and will dissociate in water. It is a very weak acid and the equilibrium vastly favours the non-dissociated form of water. The acid dissociation equilibrium of water can be written as...



With the acid dissociation equilibrium constant for water described by K_a ...

$$K_a = \frac{[H^+][HO^-]}{[H_2O]} \quad (5)$$

The K_a has been measured to be $10^{-15.7} M$.

In a dilute solution, the concentration of water does not change and is easily calculated to be $55.5 M$. Since this term does not change we can describe a new equilibrium constant, K_w , where...

$$K_w = K_a \cdot [H_2O] = K_a \cdot 55.5 M \quad (6)$$

Substituting Equation 6 into Equation 5 gives...

$$K_a = \frac{[H^+][HO^-]}{55.5 M} \quad (7)$$

and so...

$$K_w = [H^+] [HO^-] \quad (8)$$

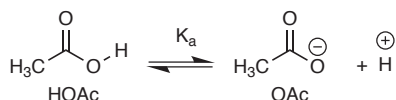
and we can calculate K_w to be $55.5 M \times 10^{-15.7} M = 10^{-14} M^2$. So we now have the famous water dissociation constant...

$$K_w = [H^+] [HO^-] = 10^{-14} M^2$$

4 Weak Acids and Bases

Most acidic groups on organic molecules are weak acids. They will not completely dissociate in water and both the acidic and basic form will exist at significant concentrations.

For example, consider acetic acid...



which we will write in an abbreviated form as...



where HA is the acidic form of the acetic acid, HOAc, which can donate the proton. A is the basic form of the acid, which can accept a proton. A is called the conjugate base of the acid, HA. In a similar manner, HA can be considered to be the conjugate acid of the base A.

A conjugate base is just the basic form of a given acid. If it accepts a proton it will become the acid.

A conjugate acid is just the acidic form of a given base. If it donates a proton it will become the base.

The equilibrium constant for a weak base is given by the equilibrium equation for the reaction described by eq. 9...

$$K_a = \frac{[H^+] [A^-]}{[HA]} \quad (10)$$

where H, A, and HA are the concentrations of H, A, and HA, respectively.

Example 1: For acetic acid we know the K_a is $10^{-4.7}M$. If I make a $50mM$ solution of acetic acid, what will the concentration of proton, the conjugate base (acetate) and the acid (acetic acid) be?

See the Equilibrium Tutorial for how to approach this problem. We start with $0.05M$ of acetic acid. Let x be the amount of acetic acid that dissociates. The concentration of acetic acid at equilibrium will be $0.05 - x$. The concentration of proton and conjugate base will both be equal to the amount of dissociation, x .

So we can write eq. 10 as...

$$10^{-4.7} = \frac{x \cdot x}{0.05 - x} \quad (11)$$

We can solve eq. 11 by rearranging it to the quadratic form and using the quadratic equation (see the Equilibrium Tutorial) or we can simply use the Maple software package to do it directly as shown in Figure 1. Maple is available on all campus computers.

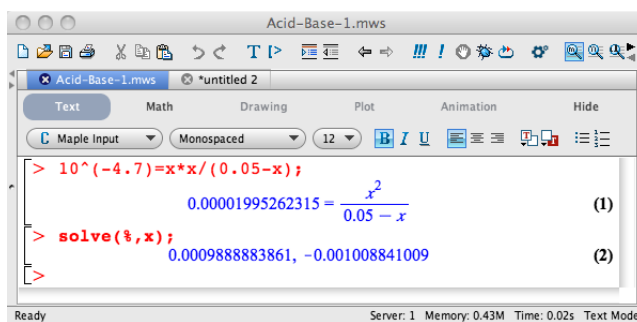


Fig. 1: Using Maple to solve a quadratic equation quickly. The % symbol means to use the most recently entered formula in the operation. x cannot be negative as that would give a negative concentration of proton and conjugate acid which is ridiculous. So we choose the root $x = 0.0009888M$.

Under the conditions in example 1, the acetic acid is only 0.0% dissociated. 99.96% is in the acidic form. Acetic acid is obviously a weak acid.

5 pH

We are always dealing with equilibrium constants on the order of 10^{-1} to 10^{-13} for weak acids. Instead of writing out exponents constantly it is useful to use a logarithmic scale. We use the p function.

$$p(x) = -\log(x) \quad (12)$$

So if we find that the concentration of proton in a solution is $10^{-3.76} M$ we can say that the $p([H^+])$ is 3.76.

$$p(10^{-3.76}) = -\log(10^{-3.76}) = 3.76 \quad (13)$$

We will write the p function of proton concentration, $p([H^+])$ as pH for convenience.

$$pH \equiv p([H^+]) \quad (14)$$

Example_2. If the pH of an acid solution is 2.38, what is the concentration of dissociated proton in the solution?

We can substitute the value for pH into equation and solve for $[H^+]$...

$$\begin{aligned} 2.38 &= -\log([H^+]) \\ -2.38 &= \log([H^+]) \\ 10^{-2.38} &= [H^+] \end{aligned}$$

So the concentration of proton is $10^{-2.38} M$ or $0.00417 M$.

We see from example 2 that a higher pH means a lower concentration of H , a lower pH means a higher $[H^+]$.

6 pKa and acids

We can use the p function on the K_a values as well. The pK_a value for acetic acid is 4.7. We see that a high pK_a value relates to a lower K_a value and therefore a weaker acid. A low pK_a value relates to a higher K_a value and therefore a stronger acid.

HCl has a pK_a value of -6 (very low) and is obviously a much stronger acid than acetic acid, which has a pK_a value of 4.7. The pK_a value for acetic acid is 10.7 log units higher than that of HCl so it is $10^{10.7}$ -fold weaker than HCl as an acid. *Convince yourself that this is true. Do the math.*

6.1 Hendersson-Hasselbach Equation

We are usually most interested in whether or not a particular acid group is in its acidic form or its basic form in a given solution with a known concentration of proton.

Example 3. What is the ratio of acetate (basic form) to acetic acid (acidic form) in a 5 mM solution of acetic acid if I arrange for the concentration of proton to be 10^{-4} M . (I added the acetic acid and then I added HCl or NaOH until the proton concentration was measured to be 10^{-4} M).

From equation 10 we know how calculate the ratio. We know the K_a and the concentration of proton. The $[A]/[HA]$ term can then be calculated.

$$\begin{aligned} K_a &= \frac{[H][A]}{[HA]} \\ \frac{K_a}{[H]} &= \frac{[A]}{[HA]} \\ \frac{10^{-4.7}}{10^{-4}} &= \frac{[A]}{[HA]} \end{aligned}$$

and so...

$$\frac{[A]}{[HA]} = 10^{-0.7} = 0.20$$

So the ratio of acetic acid to acetate is 5:1.

Deriving the Hendersson-Hasselbach Equation

We usually don't talk about proton concentration in terms of *moles/L*. We usually use *pH*. Let us apply the *p* function to both sides of equation 10 so that we can work in *pH* as our measurement of proton concentration. We will then be able to use an electronic *pH* meter to determine *pH* and use that number directly in all of our calculations.

So lets take equation 10 and apply the *p* function.

$$\begin{aligned} K_a &= \frac{[H][A]}{[HA]} \\ p(K_a) &= p\left(\frac{[H][A]}{[HA]}\right) \end{aligned}$$

p is a log function so we can separate the term on the right-hand side like so...

$$p(K_a) = p([H]) + p\left(\frac{[A]}{[HA]}\right)$$

We usually write the p function of K_a as pK_a and the p function of $[H]$ as pH . This is the Hendersson-Hasselbach equation. We can make one final small adjustment for readability, let us write out the p function for the right hand term as the $-\log$ function that it is...

$$pK_a = pH - \log \left(\frac{[A]}{[HA]} \right) \quad (15)$$

We can make rearrange the terms to solve for pH to get the more popular form of the equation...

$$pH = pK_a + \log \left(\frac{[A]}{[HA]} \right) \quad (16)$$

This is the Hendersson-Hasselbach equation in the form that we usually see it written. It is convenient because in biochemistry we almost always refer to the pK_a of an acid when we are discussing its acid equilibrium and we always refer to pH when discussing proton concentration.

Example 4. The pK_a of acetic acid is 4.7. What would the ratio of the basic to the acidic form be in a solution where the pH has been set to 7.0?

Just plug you known values into the Hendersson-Hasselbach equation...

$$\begin{aligned} pH &= pK_a + \log \left(\frac{[A]}{[HA]} \right) \\ 7 &= 4.7 + \log \left(\frac{[A]}{[HA]} \right) \\ 2.3 &= \log \left(\frac{[A]}{[HA]} \right) \\ 10^{2.3} &= \frac{[A]}{[HA]} \end{aligned}$$

The basic form (A) is in 102.3 time greater concentration than the acidic form (HA). The ratio of A:HA is 200.

Example 5. What is the percent of acetic acid that is in its dissociated, deprotonated, basic form in the case of example 4? In the case of example 3?

For example 4 we have the $[A]/[HA]$ ratio to be 200/1. That means that there are 200 molecules of acetate for every molecule of acetic acid. There are 201 molecules in total so the % of the A form would be...

$$\%A = \frac{200}{201} \times 100\% = 99.5\%$$

So acetic acid is 99.5% dissociated at pH 7.0.

For example 3 the pH is 4.0 and the ratio of $[A]/[HA]$ was calculated to be 1/5. That means that there is 1 molecules of acetate for every 5 molecules of acetic acid. There are 6 molecules in total so the % of the A form would be...

$$\%A = \frac{1}{6} \times 100\% = 16.6\%$$

another way to say this would be to say that the $[A]/[HA]$ is 0.2. So for every 0.2 molecules of the A form, there is one molecule of the HA form and therefore 1.2 molecules in total. So the % of acid in the deprotonated A form is...

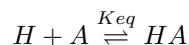
$$\%A = \frac{0.2}{1.2} \times 100\% = 16.6\%$$

7 pKa and Bases

Bases accept a proton. They do not donate a proton. So how can we use an acid dissociation constant to describe the base equilibrium for ethylamine below?

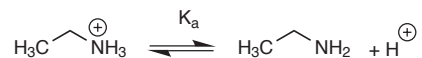


We can write this in a more general form as...

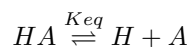


Where A is the basic form (remember that A was the basic form used in the previous cases with acid dissociation) and HA is the protonated base. HA is the conjugate acid of the base in this case.

To enable us to compare acids to bases we want to use the acid dissociation constant, K_a , to describe all acids and bases. To do this, why don't we write the reaction described above as an acid dissociation? Just look at it from the other end. Start with the acidic form (conjugate acid of the base) and have that dissociate to give the basic form and a proton.



We can now use an acid dissociation constant, K_a , to describe this equilibrium. We can now write the equilibrium in the terms of an acid dissociation...



Where HA is the acidic form of the molecule (HA is the conjugate acid of the base in this case) and A is the basic form of the molecule.

The equilibrium is then described in exactly the same manner as with acids because we are still dealing with acids (conjugate acids of bases in this case).

$$K_a = \frac{[H][A]}{[HA]}$$

Do not dwell on the conjugate acid idea too much. In all equilibria that we consider we will write the equation so that the acidic form is on the left and the basic form is on the right. That's all there is to it. The acid dissociation constant for a base will be the equilibrium of the acidic form of that base dissociating to its basic form.

As you would expect, the acidic forms of bases are weaker acids than the molecules we commonly consider to be acids. Bases prefer to accept a proton and the equilibrium will favour the acidic form over the basic form.

For example, the K_a for dissociation of the acidic form of ethylamine is $10^{-9.7}$. Compare this to the K_a for acetic acid dissociation, which is $10^{-4.7}$. We see that the K_a for acetic acid dissociation is 10^5 fold greater than that for protonated ethylamine dissociation.

Lets look at the pK_a values. Acetic acid has a pK_a of 4.7 and the acidic form of ethylamine has a pK_a of 9.7. Ethylamine is 5 log units weaker as an acid compared to acetic acid. The higher the pK_a value the weaker the acid.

Very weak acids indicate that the basic form of the acid must be a strong base since it greatly prefers to be in the protonated form. Higher pK_a values for an acid indicate that the basic form of the molecule is a stronger base.

Example 6. Construct a table of the % dissociation of acetic acid and the acidic form of ethylamine in solutions with the following pH values. 2.7, 3.7, 4.7, 5.7, 6.7, 7.7, 8.7, 9.7, 10.7 and 11.7. Draw a graph of the % dissociation vs. pH for each case.

The pK_a of acetic acid is 4.7 and the pK_a of ethylammonium is 9.7.

For a sample calculation lets take acetic acid at pH 6.7. Just plug the values into the Hendersson-Hasselbach equation.

$$\begin{aligned} pH &= pK_a + \log \left(\frac{[A]}{[HA]} \right) \\ 6.7 &= 4.7 + \log \left(\frac{[A]}{[HA]} \right) \\ 2.0 &= \log \left(\frac{[A]}{[HA]} \right) \\ 10^{2.0} &= \frac{[A]}{[HA]} \end{aligned}$$

The ratio of $[A] / [HA]$ is 100/1. There are 100 molecules of A for every molecule of HA and 101 molecules in total. So the % of A is ...

$$\%A = \frac{100}{101} \times 100\% = 99.01\%$$

You may have already noted that we can state a general formula for the % of A in the equilibrium...

$$\%A = \frac{\alpha}{\alpha + 1} \times 100\% \text{ where } \alpha = \frac{[A]}{[HA]}$$

So lets do the math and make the table...

Tab. 1: % Basic form of acetic acid and ethylamine at various pH values.

pH	Acetic Acid	Ethylamine
2.7	0.99	0.00
3.7	9.09	0.00
4.7	50.00	0.00
5.7	90.91	0.01
6.7	99.01	0.10
7.7	99.90	0.99
8.7	99.99	9.09
9.7	100.00	50.00
10.1	100.00	90.91
11.7	100.00	99.01

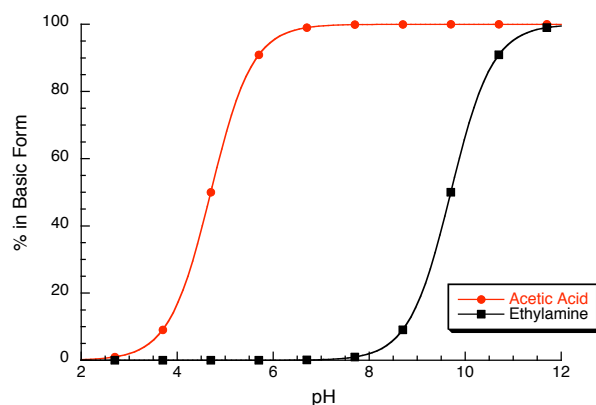


Fig. 2: Plot of the % of an acid in its basic form vs. pH of the solution.

The results in table 1 on the previous page are graphed in Figure 2 on the preceding page. I put the results for both molecules on the same graph for

comparison. pH is the variable we are controlling. Always put the variable that we are changing in the experiment on the x-axis. Put the values that we measure on the y-axis. The y-axis is the dependent variable, the variable that depends on what we are changing. The x-axis is the independent variable, the variable that we control. See the Graphing Tutorial for more information on how to make a correct graph.

The line drawn along the data points are curve fits for the function that describes the relationship between dissociation and pH which we used to generate the data. Never draw a line through your data unless it represents the best fit of a function that describes the theory for the experiment. For many experiments this would be a straight line but in this case it relates to the math discussed in example 7. NEVER CONNECT THE DOTS. See the Graphing Tutorial for more information.

Example 7 (advanced topic). What is the function to which the graphs above were fit? Derive the function from the Hendersson-Hasselbach equation.

The general formula for converting the ratio $[A] / [H]$ to % is given in example 6...

$$\%A = \frac{\alpha}{\alpha + 1} \times 100\% \text{ where } \alpha = \frac{[A]}{[HA]} \quad (17)$$

and $[A] / [HA]$ can be determined by rearranging the Hendersson-Hasselbach equation...

$$pH = pK_a + \log \frac{[A]}{[HA]}$$

to...

$$\frac{[A]}{[HA]} = 10^{(pH - pK_a)} \quad (18)$$

Substituting equation 18 into equation 17 gives...

$$\%A = \frac{10^{(pH - pK_a)}}{10^{(pH - pK_a)} + 1} \cdot 100\%$$

The y-axis is %A and the x-axis is pH so we fit the data to the function...

$$y = \frac{10^{(x - pK_a)}}{10^{(x - pK_a)} + 1} \cdot 100\% \quad (19)$$

Equation 19 was used as the curve fit for both graphs. Least squared analysis of the data fit determined pK_a to be 4.7 and 9.7 for the corresponding acids. Perfect fits like this are to be expected with perfect simulated data. You could use a curve fit like this to determine the pK_a of a weak acid in a water solution.

8 Rules of Thumb

You may have observed the following relationships as you inspected the data for example 6.

1. When the pH of a solution is equal to the pK_a value for an acid, the acid will be 50% dissociated. This is obvious from inspecting Hendersson-Hasselbach equation when we set $pH = pK_a \dots$

$$pH = pK_a + \log \frac{[A]}{[HA]}$$

and we set $pH = pK_a$, therefore...

$$\begin{aligned} 0 &= \log \frac{[A]}{[HA]} \\ 10^0 &= \frac{[A]}{[HA]} \\ 1 &= \frac{[A]}{[HA]} \\ [HA] &= [A] \end{aligned}$$

The ratio of $[A] / [HA]$ is 1 when $pH = pK_a$. The acid is 50% dissociated. *Inspect the table of calculations in example 6 and convince yourself that this is true.*

2. If the pH is higher than the pK_a , the acid is more than 50% dissociated. If the pH is less than the pK_a , the acid is less than 50% dissociated. *Inspect the table of calculations in example 6 and convince yourself that this is true.*
3. If the pH is 1 log unit higher than the pK_a value the acid is about 90% dissociated. If the pH is 2 log units higher than the pK_a value the acid is about 99% dissociated. If the pH is 3 log units higher than the pK_a value the acid is about 99.9% dissociated. And so on... *Inspect the table of calculations in example 6 and convince yourself that this is true.*
4. If the pH is 1 log unit lower than the pK_a value the acid is about 10% dissociated. If the pH is 2 log units lower than the pK_a value the acid is about 1% dissociated. If the pH is 3 log units lower than the pK_a value the acid is about 0.1% dissociated. And so on... *Inspect the table of calculations in example 6 and convince yourself that this is true.*

Example 8. A Histidine residue in a protein has a pK_a value for the acidic form of 7.0. The acidic form donates a proton during the rate-determining step of the catalytic mechanism. What is the likely effect on the rate of the enzyme-catalyzed reaction if I raise the pH from 7.0 to 8.0?

Initially, $pH = 7$ and the $pK_a = 7$ so the histidine is 50% in its acidic form. At $pH = 8.0$, which is one log unit higher than the pK_a value, we know from the rules of thumb that the histidine will be 90% in its basic form. So it is 10% in its acidic form.

We go from 50% of histidine being in the acidic form that can donate a proton and catalyze the reaction to $\sim 10\%$ being in the acidic form. This is a 5-fold drop in the concentration of enzyme able to catalyze the reaction and likely results in a 5-fold decrease in the rate of the reaction.

9 The Ultimate Rule of Thumb

Imagine two bases in a tug-o-war for a proton. In most cases that battle is between the basic form of an acid and water. Will the proton be on the acid, giving us the acidic form of the acid, or will it be on the water, giving us the protonated, acidic form of water? Who will win and by how much?

Imagine that the pK_a value of an acid represents the strength of that acid of for holding onto the proton. Now imagine that the pH value of a water solution represents the strength with which that particular solution will hold onto the proton. The stronger man wins.

- If the pK_a and the pH are the same it is a tie and 50% of the proton is on the acid and 50% is dissociated into the water solution.
- If the pH is higher than the pK_a value then the water wins and most of the proton will be dissociated from the acid. The larger the difference, the more the proportion of dissociation. See rule of thumb #3 above for more details.
- If the pH is lower than the pK_a value then the acid wins and most of the proton will be on the acid. The larger the difference, the less the proportion of dissociation. See rule of thumb #4 above for more details.

This rule can be extended to include any acid and base. Consider that I add equal amounts of acetate (basic form of acetic acid) and ethylamine (basic form of ethylamine) to a solution and then I add just one equivalent of HCl. Which base, the acetate or the ethylamine, will the excess proton just added go to? The pK_a of acetic acid is 4.7 and the pK_a of the acidic form of ethylamine is 9.7. Ethylamine wins. The difference is 5 log units so 99.999% of the proton is on the ethylamine.

A fun observation is that the number of “9”s is equal to the difference in pH log units. When the difference is 0, we have 50% dissociated. When pH is 1 unit greater we have $\sim 90\%$ dissociated; when pH is 2 units greater we have 99% dissociated; when pH is 3 unit greater we have 99.9% dissociated; when pH is

4 units greater we have 99.99% dissociated; when pH is 5 units greater we have 99.999% dissociated; and so on. . .

Example 9. In the hydrophobic core of a protein I have a carboxylic acid group from a glutamate residue and an amine group from a lysine residue. Which group will be in the acidic form and which group will be in the basic form if they must share a proton between them?

The pK_a of glutamic acid is 4.25 and the pK_a for protonated lysine is 10.53 (although they are probably different due to the non-aqueous environment in a hydrophobic protein core). So we see that the lysine wins. There is more than 6 log units in the difference so that means more than 99.9999% of the protons is on the lysine residue. (“six 9s” for 6 pH units)

Example 10. A cysteine protease uses a thiol group from a cysteine residue as a nucleophile in the rate-determining step of the catalytic mechanism. In the hydrophobic active site there is an adjacent basic histidine residue. What is the percent of thiol that is deprotonated and in its nucleophilic form?

The pK_a of an acidic histidine group is 6.0 and the pK_a of a cysteine thiol is 8.3. So the thiol wins and is at least 99% in its acidic form. Which means that it is just under 1% in its basic, nucleophilic form. This is apparently enough as the enzyme is active. Other factors in the active site may contribute to boost the amount of nucleophilic form.

We can use equilibrium math to determine the exact ratio of acidic forms to be 199.5:1 for cysteine:histidine. This corresponds to 99.50% of the cysteine being protonated with 0.50% in the basic, nucleophilic form.

Part II. Supplemental Advanced Topic: Simulating pH titrations

10 Introduction

Suppose that you want to create a data set for a titration to construct a graph. You may wish to show what a titration curve would look like for various compounds for teaching purposes or to impress your friends. If you are curious how you can simulate a pH titration for a molecule mathematically I will reveal all my secrets here. Figure 3 shows a titration curve calculated for some amino acids with ionizable side chains.

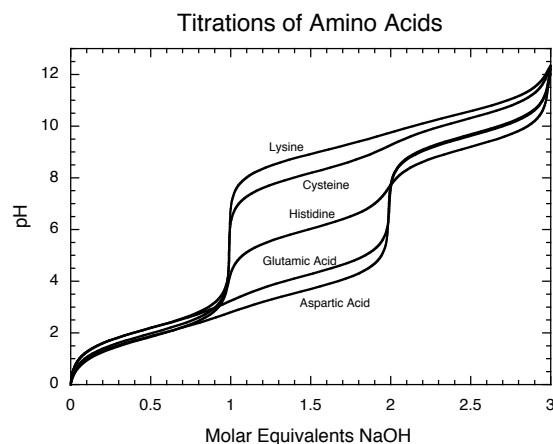


Fig. 3: Titration of amino acids with three acidic groups.

The titration experiment above was performed as follows: The amino acid was dissolved to give a known concentration. Enough HCl was added to lower the pH to 0. We then began adding a NaOH solution and measuring the pH at each step until we reached a pH of 12. We plot the measured pH vs. molar equivalents of NaOH (If we have 5 mM of amino acid and I add 10 mM of base, that is 2 equivalents).

The above graph has been corrected for the effect of water acid/base equilibrium and assumes that the volume does not change during the titration. We can correct for the titration of water by subtracting the data set for a titration of pure water. We can assume that the volume does not change if we use a large volume of solution and the titrant is very concentrated so that there is only a very small change in the volume over the titration. We can accomplish this correction in the calculation by ignoring the acid equilibrium of the water solution and any volume changes.

So, how did I calculate these graphs (and others that you will see in this course)?

11 Simulating a Titration

Let us consider the simple case of a molecule with a single acidic group. The situation can be described by the reaction of equation 20 and the acid dissociation constant is described by equation 21.



$$K_a = \frac{[H][A]}{[HA]} \quad (21)$$

To simulate our experiment we want to know how much base to add to take us from the pH where we start the titration (in this case, 0) to the pH where we are measuring the amount of base that had been added.

So let $[B]$ be the concentration of NaOH added. The amount of base added would be equal to the concentration of OH in the solution and the concentration of any deprotonated molecules (which had given up a proton to the strong hydroxide base).

$$[B] = [OH] + [A] \quad (22)$$

Where $[OH]$ is the concentration of hydroxide in the solution and $[A]$ is the concentration of the basic form of the molecule.

We do not know the concentration of A but we do know the total concentration of acidic molecule, $[A]_t$.

$$[A]_t = [A] + [AH] \quad (23)$$

We can find $[A]$ in terms of $[A]_t$ by multiplying $[A]_t$ by the mole fraction of $[A]$ in the solution.

$$[A] = [A]_t \cdot \frac{[A]}{[A]_t} \quad (24)$$

Substituting equation 23 into equation 24 gives...

$$[A] = [A]_t \cdot \frac{[A]}{[A] + [AH]} \quad (25)$$

Substituting equation 21 into equation 25 gives...

$$[A] = [A]_t \cdot \frac{K_a \cdot \frac{[HA]}{[H]}}{K_a \cdot \frac{[HA]}{[H]} + [AH]} \quad (26)$$

Simplify by multiplying the equation by $[H] / [HA]$ in both the numerator and denominator...

$$[A] = [A]_t \cdot \frac{K_a}{K_a + [H]} \quad (27)$$

Substituting equation 27 into equation 22 gives...

$$B = [OH] + [A]_t \cdot \frac{K_a}{K_a + [H]} \quad (28)$$

We know how to convert concentration of OH in water to concentration of H from equation 8...

$$K_w = [H][OH] = 10^{-14} M^2$$

Substituting equation 8 into equation 28 gives the function that will describe the titration curve in terms of base added vs. $[H]$.

$$B = \frac{10^{-14}}{[H]} + [A]_t \cdot \frac{K_a}{K_a + [H]} \quad (29)$$

We could also have the function with pH and pK_a instead of $[H]$ and K_a as the variable by substituting equation 14 into equation 29...

$$B = \frac{10^{-14}}{10^{-pH}} + [A]_t \cdot \frac{10^{-pK_a}}{10^{-pK_a} + 10^{-pH}} \quad (30)$$

Let us say the pK_a of the acid is 5.3 and we have added enough HCl to make sure that all of the molecules were in the acidic form, HA. We begin adding NaOH solution and start recording pH at $pH = 0$ until $pH = 12$.

Note that the math we are using uses pH as the variable to calculate the amount of hydroxide needed to reach that pH from a fully protonated condition. So in this calculation we are treating pH as the variable and calculating hydroxide added when in a titration we use hydroxide added as the variable and measure pH . The math is just easier this way. Eq. 29 is nice and elegant when solving for $[B]$ in terms of $[H]$. It becomes very complicated quadratic equation if we wanted to rearrange it to solve for $[H]$ in terms of $[B]$.

We will create a set of data from a set of pH values and then plot pH vs. hydroxide added in the correct manner for a titration. We will use equation 30 and a set of pH values. We will set $[A]_t = 1$ for convenience and the pK_a value is 5.3. The data is presented in Table 2 on the following page and plotted in Figure 4 on the next page. The curve fit is the least squares best fit for the function used to create the data. You will have to assign the pH values as the x-axis in the curve fit program for computational reasons discussed above.

Tab. 2: Calculated titration data using Eq. 30. Equivalents of base added to reach given pH value are reported.

pH	Equiv. Base Added
0	0.000
1	0.000
2	0.001
3	0.005
4	0.048
5	0.334
6	0.834
7	0.980
8	0.998
9	1.000
10	1.000

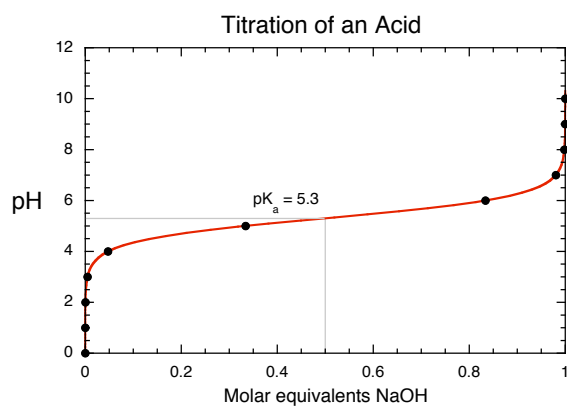
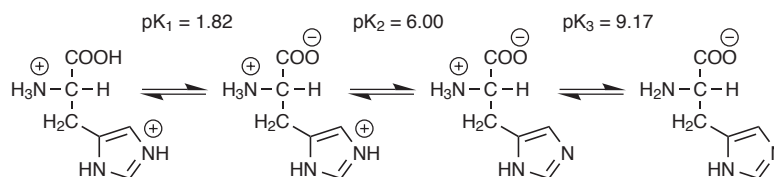


Fig. 4: Simulated titration of a simple acid with a pK_a of 5.3. Data is from Table 2 and the curve fit is Eq. 30.

12 Simulating a More Complex System

What about an amino acid with three ionizable groups? Lets take histidine with a pK_a value for the α -COOH group of 1.82, pK_a value for the acidic form of the α -NH₂ group of 9.17, and a pK_a value for the acidic form of the imidazole side group of 6.00.



This can be written in more general terms as...



And the total concentration of $[A]$ is given by...

$$[A]_T = [H_3A] + [H_2A] + [HA] + [A] \quad (34)$$

The respective K_a values will be called K_1 , K_2 , K_3 . They are calculated as...

$$K_1 = \frac{[H_2A][H]}{[H_3A]} \quad (35)$$

$$K_2 = \frac{[HA][H]}{[H_2A]} \quad (36)$$

$$K_3 = \frac{[A][H]}{[HA]} \quad (37)$$

We will start with a system that is 100% $[H_3A]$ and add hydroxide until the pH reaches 12.0. the amount of hydroxide added will be the sum of the concentration of hydroxide and the concentrations of the deprotonated species as shown in equation 38.

$$[B]_T = [HO^-] + [H_2A] + 2 \cdot [HA] + 3 \cdot [A] \quad (38)$$

2 hydroxides were required to titrate $[H_3A]$ to $[HA]$ and 3 were required to titrate $[H_3A]$ to $[A]$.

We can find a term for $[H_2A]$ just like in the simple case...

$$\begin{aligned} [H_2A] &= [A]_T \frac{[H_2A]}{[A]_T} \\ [H_2A] &= [A]_T \frac{[H_2A]}{[H_3A] + [H_2A] + [HA] + [A]} \end{aligned} \quad (39)$$

Let us convert all of the terms so that they contain only the concentration of $[H]$ and $[H_3A]$ as variables. We can rearrange equation 35 to find $[H_2A]$.

$$[H_2A] = [H_3A] \frac{K_1}{[H^+]} \quad (40)$$

So by substituting equation 40 into equation 39, we now have...

$$[H_2A] = [A]_T \frac{[H_3A] \frac{K_1}{[H^+]}}{[H_3A] + [H_3A] \frac{K_1}{[H^+]} + [HA] + [A]} \quad (41)$$

And from equation 36 we see that...

$$[HA] = [H_2A] \frac{K_2}{[H^+]} \quad (42)$$

and so with substituting equation 40 into equation 42...

$$[HA] = [H_3A] \frac{K_1}{[H^+]} \frac{K_2}{[H^+]} \quad (43)$$

In a similar manner we find that...

$$[A] = [H_3A] \frac{K_1}{[H^+]} \frac{K_2}{[H^+]} \frac{K_3}{[H^+]} \quad (44)$$

Substituting equations 40, 43 and 44 into equation 39 gives...

$$[H_2A] = [A]_T \frac{[H_3A] \frac{K_1}{[H^+]}}{[H_3A] + [H_3A] \frac{K_1}{[H^+]} + [H_3A] \frac{K_1}{[H^+]} \frac{K_2}{[H^+]} + [H_3A] \frac{K_1}{[H^+]} \frac{K_2}{[H^+]} \frac{K_3}{[H^+]}} \quad (45)$$

Which simplifies to...

$$[H_2A] = [A]_T \frac{\frac{K_1}{[H^+]}}{1 + \frac{K_1}{[H^+]} + \frac{K_1 K_2}{[H^+]^2} + \frac{K_1 K_2 K_3}{[H^+]^3}}$$

Which is further simplified to...

$$[H_2A] = [A]_T \frac{K_1 [H^+]^2}{[H^+]^3 + K_1 [H^+]^2 + K_1 K_2 [H^+] + K_1 K_2 K_3} \quad (46)$$

In a similar manner we find the equations for $[HA]$ and $[A]$ in terms of $[H]$...

$$[HA] = [A]_T \frac{K_1 K_2 [H^+]}{[H^+]^3 + K_1 [H^+]^2 + K_1 K_2 [H^+] + K_1 K_2 K_3} \quad (47)$$

$$[A] = [A]_T \frac{K_1 K_2 K_3}{[H^+]^3 + K_1 [H^+]^2 + K_1 K_2 [H^+] + K_1 K_2 K_3} \quad (48)$$

So we now can substitute equations 46, 47 and 48 into equation 38 to give...

$$[B] = \frac{K_w}{[H^+]} + [A]_T \frac{K_1 [H^+]^2 + 2K_1 K_2 [H^+] + 3K_1 K_2 K_3}{[H^+]^3 + K_1 [H^+]^2 + K_1 K_2 [H^+] + K_1 K_2 K_3} \quad (49)$$

We can solve it for pH and pK_a values as well...

$$[B] = \frac{K_w}{10^{-pH}} + [A]_T \frac{10^{-(pK_1+2 \cdot pH)} + 2 \cdot 10^{-(pK_1+pK_2+pH)} + 3 \cdot 10^{-(pK_1+pK_2+pK_3)}}{10^{-3 \cdot pH} + 10^{-(pK_1+2 \cdot pH)} + 10^{-(pK_1+pK_2+pH)} + 10^{-(pK_1+pK_2+pK_3)}} \quad (50)$$

Equation 50 describes the titration curve for a triprotic acid in terms of pH and pK_a .

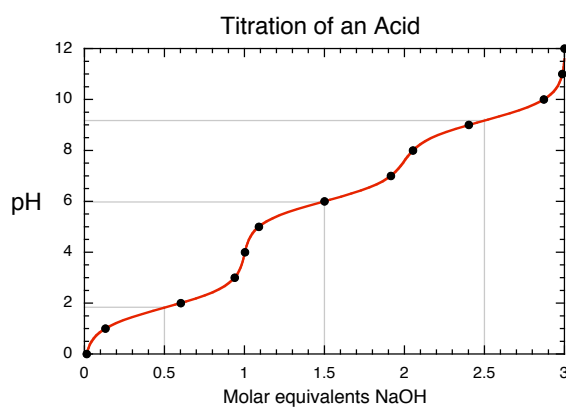
Now lets see how much NaOH solution we must add to raise the pH to the following pH values. Table 3 on the following page shows the results of the calculation using equation 49 and the pK_a values for histidine.

The data from table 3 has been plotted in figure 5 on the next page and a curve fit using the function described by equation 50 has been applied using the pH data set as the x-axis and the calculated data set for moles of base added as the y-axis. We have to do the curve fit this way because of the fact that the function solves for $[B]$ using pH as the variable. Solving the function the other way around is a heap of trouble.

In an experiment we would be changing the amount of base added and measuring the pH . So base added is the x-axis (independent variable) and the measured pH is the y-axis (dependent variable). This is the way the experimental data is plotted so that is the way we plotted the calculated data set. See the Graphing Tutorial for how to decide which axis to put each data set upon.

Tab. 3: Calculated titration data for triprotic acid.

pH	Base Added
0	0.00000
1	0.00001
2	0.00010
3	0.00100
4	0.0101
5	0.111
6	1.500
7	2.889
8	2.989
9	2.999
10	3.000
11	3.001
12	3.010

Fig. 5: Simulated titration of a triprotic acid with a pK_1 of 1.8, pK_2 of 6.0, and pK_3 of 9.16.

12.1 Now it's your turn.

Derive the formula for describing the titration curve of a molecule with two acidic groups.

Your answer should be...

$$[B] = \frac{K_w}{[H^+]} + [A]_T \frac{K_1 [H^+] + 2K_1 K_2}{[H^+]^2 + K_1 [H^+] + K_1 K_2} \quad (51)$$